The empirical recurrence for OEIS A282789: recovering a conjecture that is not written down

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Abstract

OEIS A282789 records "Empirical recurrence of order 60", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 125$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 60, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 60 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282789 is "Number of nX6 0..1 arrays with no 1 equal to more than one of its king-move neighbors, with the exception of exactly one element.". It has offset 1 and begins

$$12, 240, 8216, 119176, 2207352, 34974844, 545174028, 8385651160, \ \ldots$$

The entry states:

> Empirical recurrence of order 60 (see link above)

As of the "Last modified" line on the live entry (Feb 21 2017 08:44:13, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1370$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 125$ of the 1370, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 125.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 250$ exact terms of the model returns order 60 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{60} c_i\, a(n-i),$$

c_1	8	c_{16}	−95790529305	c_{31}	−41809888481640	c_{46}	349943558874556
c_2	142	c_{17}	368960698772	c_{32}	−502023438595345	c_{47}	81711212718806
c_3	582	c_{18}	−1049801919266	c_{33}	−101454283839846	c_{48}	−156816034830829
c_4	−10565	c_{19}	1644502273816	c_{34}	150774816700059	c_{49}	−101712923683384
c_5	−103130	c_{20}	−4558155871905	c_{35}	760821910317746	c_{50}	30291062186053
c_6	−443213	c_{21}	8520110236862	c_{36}	314780075186914	c_{51}	52205768281808
c_7	360238	c_{22}	−9561859464170	c_{37}	−267918248360742	c_{52}	6534399479889
c_8	6598770	c_{23}	25852310503990	c_{38}	−873695827469142	c_{53}	−14259846313340
c_9	31037506	c_{24}	−33740178456168	c_{39}	−346826400229822	c_{54}	−6143201890015
c_{10}	−13042102	c_{25}	24677542365950	c_{40}	436961419196245	c_{55}	1448181627582
c_{11}	−46887694	c_{26}	−93912178515770	c_{41}	774042761721252	c_{56}	1480604897034
c_{12}	−1239673350	c_{27}	72745528782022	c_{42}	97246232739434	c_{57}	147151141020
c_{13}	962146338	c_{28}	−20184641617835	c_{43}	−532421135483670	c_{58}	−109329023673
c_{14}	−8075207667	c_{29}	246905186644856	c_{44}	−430915259409782	c_{59}	−31165618968
c_{15}	50842109548	c_{30}	−57352992694403	c_{45}	156400988011084	c_{60}	−2372469264

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 60, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $60+4$ terms removed returns order 60 as well, so $\deg q_{\min} = 60 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $60+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 60 that this sequence satisfies — and that family turns out to have exactly one member.

References

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